Quant Strategy Specification & Operations Manual

Multi-Tier Architecture, Backtest Analytics & Live Execution Workflow | Equity Pipeline

1. MULTI-TIER SYSTEM ARCHITECTURE

The strategy operates across a modular 6-layer architecture separating data ingestion, macro regime evaluation, signal generation, risk management, and narrative filtering:

Layer	Directory / Module	Functional Gate	Core Mechanism / Rule
Layer 0	`src/data/data_quality.py`	Data Integrity & Sanity	Handles split adjustments, missing bars, and point-in-time EOD price cache.
Layer 1	`src/regimes/macro_filter.py`	Macro & Regime Engine	Evaluates SPY trend regime to set risk sizing multipliers (1.0, 0.5, 0.0).
Layer 2	`src/confirmation/relative_strength.py`	Universe Gating & RS	Ranks universe on 63-day excess ROC vs. SPY; gates top 50%–55% leaders.
Layer 3	`src/setups/demand_zones.py`	Demand Zone & Volume Setups	Flags breakouts with volume exceeding $\text{Mean}_{20} + 1.5 \times \text{StdDev}_{20}$.
Layer 4	`src/risk/exposure_manager.py`	Position Sizing & Exposure	Applies flat $1,000 risk model, single 1% stop buffer, and 60-day time-stop.
Layer 5	`src/narrative/catalyst_filter.py`	Narrative & Catalyst Filter	Blocks setup entries if earnings or major news is scheduled within 7 days.

2. TWO-PHASE OPERATIONAL WORKFLOW

PHASE A **Parameter Optimization (Offline / Weekly)**

Execute `cache_manager`, `engine`, and `analyze_results` to sweep relative strength percentiles (0.05 to 0.80). Determines the optimal threshold (e.g., `top_pct = 0.50` or `0.55`) maximizing net PnL and profit-per-trade.

PHASE B **Daily Setup Generation (Post Market Close)**

Update the daily EOD cache post-close, then run `demand_zones.py` using the optimal `PCT_PARAM`. Inspect individual stock win/loss metrics and daily supply zones prior to routing orders for next-day open.

3. MATHEMATICAL RISK & EXECUTION FORMULAS

Stop Loss (Single Buffer): $\text{Stop Price} = \text{round}(\text{Low}_{T-1} \times 0.99, 2)$

Take-Profit Target (2:1 Reward-to-Risk): $\text{Target} = \text{round}(\text{Open}_{T+1} + [2.0 \times (\text{Open}_{T+1} - \text{Stop Price})], 2)$

Position Sizing: $\text{Shares} = \text{floor}((\$1{,}000 \times \text{Regime Multiplier}) / (\text{Open}_{T+1} - \text{Stop Price}))$

4. PARAMETER SWEEP ANALYTICS & ROBUSTNESS CURVES

Historical parameter sweeps evaluate system trajectory across relative strength filter levels (5% to 80%). The primary evaluation metrics confirm system robustness against parameter overfitting:

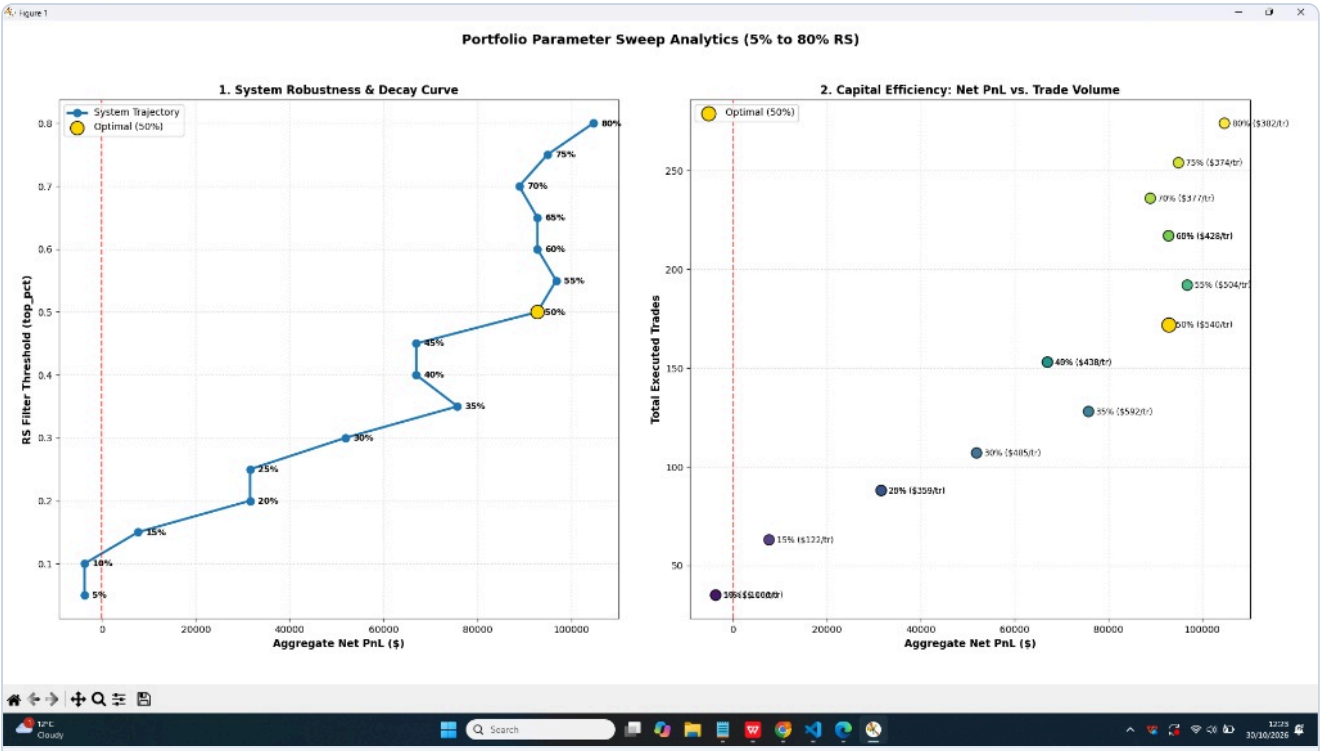


Figure 1: System Robustness Curve (Left) showing PnL expansion up to 50%–55% RS thresholds, and Capital Efficiency Plot (Right) illustrating trade volume vs. net profit.

Backtest Sweep Benchmarks (2024–2026 Dataset) — top_pct = 0.55 Benchmark

$100,959.14	$82,323.80	49.2% / 51.7%	193 / 172	~3.0x
MAX NET PNL (55%)	60D TIME STOP (50%)	WIN RATES	TOTAL TRADES	PAYOFF MULTIPLE

Note: While the strategy achieves a ~50% win rate, its expectancy is driven by a ~3:1 payoff ratio where winning trades yield three times the average dollar loss. Trailing stops were excluded from backtests and are designated for live execution testing.

5. INTRADAY SLIPPAGE AUDIT & ASSET EQUITY PATH FILTERING

To eliminate backtest execution reporting bias, `fetch_intraday_data.py` downloads actual 1-minute, 5-minute, and 10-minute prices on execution dates. `add_intraday_checks.py` stress-tests trades against 1-minute fills (`price_1m`) and analyzes winning trade tail distributions (IQR):



Figure 2: QCOM Individual Equity Path ($9,984.96 Net PnL, 75% Win Rate). Green dots denote wins, red dots represent losses.

1-Min Execution Distribution

- **Audited Net PnL:** $90,083.48
- **Audited Win Rate:** 55.8% (96/172)
- **Winning Trade Q1:** $1,295.78
- **Winning Trade Q3:** $2,051.10
- **Interquartile Range:** $755.32
- **Core Trades in IQR:** 48 Wins
- **Extreme Winner Outliers:** 3 Trades

Local Regime Shift Protocol: Prior to taking a live setup, the trader inspects the asset's individual equity path. If a stock displays a recent cluster of consecutive losses (red dots), it signals local regime decay or choppy distribution—the trade is skipped even if overall long-term PnL is positive.

6. SCRIPT REFERENCE & PRE-TRADE APPROVAL CHECKLIST

```
# 1. Activate VM Environment: . venv/Scripts/Activate.ps1 # 2. Update EOD Cache : python -m src.data.cache_manager # 3. Daily Setup Scanner : python -m src.setups.demand_zones # 4. Parameter Backtest : python -m src.backtest.engine # 5. Sweep Analytics Graph : python -m src.backtest.analyze_results # 6. Fetch Intraday Ticks : python fetch_intraday_data.py # 7. Stress-Test Slippage : python add_intraday_checks.py
```

Step	Verification Gate	Validation Criteria	Failing Action
1	Relative Strength Rank	Asset ranks in Top 50%–55% excess 63-day ROC vs. SPY	SKIP SETUP
2	Volume Spike Threshold	Day T volume > $\text{Mean}_{20} + 1.5 \times \text{StdDev}_{20}$	SKIP SETUP
3	Catalyst News Filter	No earnings or major news events scheduled in next 7 days	BLOCK TRADE
4	Equity Path & Loss Cluster	No recent cluster of consecutive losses; positive asset curve	SKIP / REGIME DECAY
5	Discretionary Supply Check	Chart inspection shows clear headroom to major supply zones	REJECT ENTRY

3. TWO-PHASE OPERATIONAL WORKFLOW

To run the system effectively, the operator follows two clear sequential phases:

- **PHASE A — Historical Optimization & Parameter Calibration:**
  - **Step 1:** Update cached price/volume data (`python -m src.data.cache_manager`).
  - **Step 2:** Run the backtest sweep engine (`python -m src.backtest.engine`) across Relative Strength thresholds.
  - **Step 3:** Run analytics (`python -m src.backtest.analyze_results`) to determine the optimal RS parameter (e.g., `top_pct = 0.55` or `0.50`).
- **PHASE B — Daily Post-Close Live Setup Generation:**
  - **Step 1 (Mandatory Daily):** Run `data.cache_manager` immediately after US market close to ingest the latest daily bar data.
  - **Step 2:** Set PCT_PARAM in setup configuration to match the optimal RS threshold identified in Phase A.
  - **Step 3:** Run setup scanner (`python -m src.setups.demand_zones`) to output actionable buy candidates for next day's open.
  - **Step 4:** Manually inspect the individual ticker tab in `sweep_analytics.xlsx` to verify historical win/loss performance and check for recent loss clusters (regime shift indicator).

4. SCRIPT EXECUTION & TOOL REFERENCE

- `. venv/Scripts/Activate.ps1` — **Virtual Environment:** Activates isolated Python virtual environment.
- `python -m src.data.cache_manager` — **Data Ingestion:** Refreshes daily OHLCV price/volume caches post-close.
- `python -m src.setups.demand_zones` — **Setup Scanner:** Generates post-market buy setups for next day open.
- `python -m src.backtest.engine` — **Backtest Engine:** Sweeps RS parameters and generates asset equity curves.
- `python -m src.backtest.analyze_results` — **Analytics Module:** Plots decay curves and identifies optimal RS thresholds.
- `python fetch_intraday_data.py` — **Microstructure Audit:** Downloads 1m, 5m, and 10m tick data for backtested entry dates.
- `python add_intraday_checks.py` — **Slippage Stress-Test:** Injects 1m prices into Excel, applies conditional formatting (blue=below entry, red=above entry), recalculates audited 1m Net PnL, and outputs win/loss distribution statistics.

Capital-Efficient Threshold Metrics (Sweep Results) — top_pct = 0.55 (55%)

$100,959.14	49.2%	193	5.85	~3:1
AGGREGATE NET PNL	OVERALL WIN RATE	TOTAL TRADES	AVG TRADES / MO	PAYOFF MULTIPLE

5. SUMMARY CHECKLIST FOR SETUP APPROVAL